

Figure 4. Weight storage, $3.41 \times$ smaller.

Built and verified from the real checkpoint on one NVIDIA H100. Planned and actual byte totals agree exactly. Symmetric signed INT4 with group size 32 on the reduction axis and BF16 per-group scales, which is 4.5 bits per parameter once the scales are counted. Source: engine/quant/QUANTIZATION-RESULTS.md

